SENTRA
BY CLOVE TECHNOLOGIES

SENSOR SELECTION GUIDE

Which sensor to use for strain, vibration, tilt, displacement, settlement and fatigue: start from the failure mode you are worried about, and match it to the measurement that captures it.

START FROM THE FAILURE MODE

Match the question to the measurement. Every sensor here is one we supply and install; figures come from manufacturer datasheets.

FAILURE MODE	MEASURE	WITH	SAMPLING
Fatigue in steel members	Dynamic strain, stress ranges, cycle counts	Strain gauges on critical details + synchronised DAQ	50–100 Hz under traffic
Loss of stiffness	Natural frequency and damping over time	Accelerometers, plus temperature	100–200 Hz for modal work
Settlement or scour	Rotation and vertical movement of supports	Tiltmeters and GNSS	Periodic, with motion-triggered readings
Deflection under load	Mid-span displacement	LVDTs or laser distance	Synchronised with strain during load tests
Tunnel or excavation deformation	Convergence and crown settlement	Laser tiltmeters	Periodic, per excavation stage

FAILURE MODE	MEASURE	WITH	SAMPLING
Construction or blast vibration	Peak particle velocity and frequency	Vibration meters	1000 Hz internal; reports every 10 min–6 h or on exceedance
Crack and joint movement	Opening, closing and growth of known cracks	Vibrating wire crackmeters via the G7 logger	Periodic
Pore pressure and water level	Groundwater and uplift pressure	Vibrating wire piezometers via the G7 logger	Periodic

Not in our range: fibre Bragg grating sensing and corrosion sensing.

THE SENSORS

Key datasheet figures for each sensor, and what an engineer reads from it.

Strain gauge

Vibrating wire, models 4100 / 4150

3,000–10,000 microstrain range

±0.5% accuracy (datasheet)

IP67 stainless steel

Surface strain; stress via the elastic modulus. Spot-welded or epoxy/weld mounted on steel; also concrete and rock.

Accelerometer

Triaxial MEMS, model 4030

±2 / ±6 g range options

0.5–250 Hz bandwidth

IP65 –40 to +80 °C

Vibration and dynamic response in three axes. Engineers read natural frequencies, damping and peak acceleration.

Vibration meter

Worldsensing, wireless

1000 Hz internal sampling
0–60 mm/s PPV (±0.01)
IP68 up to 2 yr battery

Peak particle velocity and dominant frequency, processed on board. Bolted to structure or ground; no cable run.

Tiltmeter

Worldsensing, wireless

±0.005° accuracy (XHP)
±0.0001° resolution
10 yr battery, IP68

Inclination in three axes. Pier and support rotation, the earliest sign of settlement. LoRaWAN, up to 15 km in open terrain.

Laser tiltmeter

Worldsensing, wireless

±1 mm at 10 m
0.1 mm resolution
150 m max range

Distance to a fixed target plus 3-axis tilt. Tunnel convergence and crown settlement, with no contact with the moving surface.

GNSS meter

Worldsensing, RTK multi-constellation

2 mm horizontal, 24 h aggregate
184 channels
IP68 −40 to +80 °C

3D position of a point over time. Long-term settlement and displacement, with a base unit on stable ground.

LOGGERS AND GATEWAYS

How readings leave the sensor. Strain, LVDT and accelerometer channels can share one clock at 100 Hz for load tests and dynamic response.

G7 vibrating wire logger

1, 2 or 6 channels; reads any major VW sensor brand. LoRaWAN or 4G.

Analog data logger

4 channels: voltage, current, Wheatstone bridge, thermistor, for LVDTs and load cells.

Digital data logger

RS485 Modbus RTU sensors onto the wireless network.

4G gateway

Up to 1000+ nodes per network, 4G LTE backhaul, IP68.

Repeater

Extends LoRaWAN underground and around obstacles.

FREQUENTLY ASKED QUESTIONS

Which sensors are used for bridge monitoring?

Typically strain gauges on girders and deck sections for stress and fatigue, tiltmeters on piers and bearings for rotation and settlement, and accelerometers for natural frequency and dynamic response. GNSS covers long-term displacement, and LVDTs measure deflection during load tests.

What sampling rate does structural monitoring need?

Static parameters such as settlement, tilt and crack width need only periodic readings. Dynamic strain under traffic is typically sampled at 50–100 Hz, and bridge modal analysis at 100–200 Hz, several times the highest frequency of interest.

Should I choose wired or wireless sensors?

Wireless LoRaWAN sensors avoid cable runs and suit long or hard-to-reach structures, with repeaters underground or around obstacles. Wired sensors such as vibrating wire gauges, LVDTs and Modbus devices join the same wireless network through G7, analog and digital data loggers.